

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
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Cloud and Aerosol Detection Software

Version 3

Top Level Design

version 1.2
06/07/2020

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ECMWF

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Change record			
Version	Date	Author / changed by	Remarks
1.0	01/04/2020	Reima Eresmaa	Initial version
1.1	07/04/2020	Reima Eresmaa	More specific on input header information
1.2	06/07/2020	Reima Eresmaa	Clarify on the concept and use of channel height assignments. Emphasize the need of external software to provide input data.

The EUMETSAT Network of Satellite Application Facilities	 NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
---	--	--	--

TABLE OF CONTENTS

1 INTRODUCTION 3

2 METHOD 3

2.1 Cloud detection..... 3

2.2 Aerosol detection..... 4

2.3 Trace-gas detection 4

2.4 Detection of land sensitivity 5

3 MODULE DESIGN..... 5

3.1 Input..... 5

3.2 Functional elements 6

4 FUNCTIONAL REQUIREMENTS..... 8

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
---	---	---

1 INTRODUCTION

The Cloud and Aerosol Detection Software (CADS) is a flexible stand-alone cloud and aerosol detection package intended for radiance assimilation studies using high-spectral-resolution infrared (IR) sounders. The software determines those IR sounder channels that (at each location) can be considered usable for the assimilation. The usability may be limited due to presence of various substances in the sounder's field-of-view (FOV). Most typically, such limitations occur due to ice- and/or liquid-phase clouds, but aerosol and excessive trace-gas concentration are also considered as potential contaminants. Furthermore, when IR sounder observations are made over land areas, the software identifies channels that may be overly sensitive to the underlying surface and related inaccuracies in the modelling of surface emission.

New functionalities in version 3.1 of CADS are added in response to the user requirement review, as documented in the CADS V3 Product Specification. The new functionalities are listed as follows:

1. Distinguishing between Saharan dust, volcanic ash and other aerosol
2. Channel-dependent flagging depending on the aerosol type
3. Situation-dependent identification of channels sensitive to underlying land surface
4. Detecting and flagging data affected by excessive Hydrogen Cyanide (HCN)
5. Processing capability for IKFS-2 (Infrared Fourier Spectrometer -2) sounder

2 METHOD

The software package consists of four modules that are independent of each other. The four modules correspond to the detection of cloud, aerosol, trace gas, and land sensitivity.

2.1 Cloud detection

Channels contaminated by cloud are inferred from brightness temperature (T_B) departures between the observed spectrum and clear-sky values that correspond to a background estimate of the atmospheric state. Most typically, the clear-sky T_B are calculated by applying a radiative transfer model on short-range NWP forecast profile data. Using channel height assignments¹ as calculated from the output of a radiative transfer model, the T_B departures are re-ordered in a vertically ranked space such that the characteristic signature of cloud becomes monotonic and more readily identifiable. A smoothing filter is applied to the ranked departures to reduce the effect of instrument noise. One of three possible scenarios is chosen depending on properties of the vertically ranked and smoothed departures. The height level (or channel index in the ranked space), where cloud contamination becomes identifiable, is determined with the help of pre-determined threshold parameter values.

¹ Channel height assignment indicates a level where a hypothetical opaque cloud top would alter the radiance measurement exactly by a certain fixed percentage (e.g. 1%) from the clear-sky value. Channel height assignments are mandatory input and they are made use of at several modules.

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
---	---	--

In addition to the observed minus calculated T_B departures, it is possible to use statistical T_B information in clusters of collocated imager pixels to assist the identification of cloudy scenes. Provision of such additional data is optional and will not serve as a substitute for the observed and calculated T_B data on IR sounder channels.

The cloud detection may be done individually for any number of spectral bands (e.g., the $15\mu\text{m}$ CO_2 band, $6\mu\text{m}$ H_2O , and $4.2\mu\text{m}$ CO_2 band may each be treated separately). Alternatively, the diagnosed cloud top height assignment in any one band can be used to infer cloud-contaminated channels in any other band.

2.2 Aerosol detection

Aerosol is detected from inter-channel differences in observed T_B in a way that exploits spectral signals unique to aerosol. Specifically, the detection scheme assumes the greatest aerosol effect to occur in the atmospheric window region near $10\mu\text{m}$, and this to be identifiable when compared with window channels near $8\mu\text{m}$. Once aerosol is found to affect the T_B observation, it is classified as either volcanic ash, Saharan dust, or other aerosol. The classification depends on further checks on inter-channel T_B differences against pre-determined thresholds. Positive aerosol detection translates to raising aerosol flags according to the following rules:

1. For volcanic ash, the positive detection means that all channels are flagged as affected by aerosol.
2. For Saharan dust, the vertical extent of aerosol is quantified to allow flagging only those channels that are affected at each location. First, Aerosol Optical Depth is estimated from the observed inter-channel T_B differences. An empirically determined function is then applied to convert the AOD estimate into a flagging threshold. Finally, channels to be flagged are determined by comparing the channel height assignments with the threshold.
3. For other aerosol, stratospheric channels are considered free from contamination, while all tropospheric-sensitive channels are flagged. The separation between stratospheric- and tropospheric-sensitive channels is made using the channels' height assignment information only.

2.3 Trace-gas detection

For the detection of excessive trace gas concentration, two groups of channels are formed. Firstly, a group of tracer channels is identified where sensitivity to the given trace gas is high. Secondly, a group of control channels is identified such that the response to temperature and humidity perturbations is similar to that of the tracer channels, but the sensitivity to the trace gas is low. Both observed T_B and the observed minus calculated departure are averaged in the two groups, and tracer minus control differences are calculated. Excessive trace gas concentration manifests itself by the two differences falling below pre-determined threshold values, in which case the trace-gas contamination flags are raised for those channels known to hit the absorption lines of this gas.

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
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2.4 Detection of land sensitivity

Detection of land-sensitivity relies on normalized channel height assignments. The normalization is done separately at each location such that the highest- and lowest-ranked channels are given values 0 and 1, respectively. Two tuning parameters are specified: firstly, a threshold is set to determine which fraction of land (inside the sounder FOV) activates the land-sensitivity detection. Secondly, a threshold is set for the normalized height assignment to identify the surface-sensitive channels. Channel is flagged land-sensitive when both the input land fraction and the normalized height assignment exceed their respective threshold values.

By setting the land fraction threshold to zero, this functionality generalizes to detecting surface-sensitive channels over any surface type (i.e., not just land, but also sea and sea ice).

3 MODULE DESIGN

3.1 Input

Mandatory input to the software consists of a list of included sounder channels, observed and calculated (clear sky) T_B , channel height assignments, and header information such as the land fraction and height assignments that correspond to the tropopause and the top of planetary boundary layer at the sounder FOV location. The calculated T_B should correspond to an assumption of neither cloud, aerosol nor excessive trace gas in the FOV.

Optional input consists of a namelist file that may be used to adjust the settings and channel selections used throughout the running of the code, a list of collocated imager channels to be used, statistical information of T_B observation in collocated imager pixels, and calculated clear-sky T_B on each imager channel.

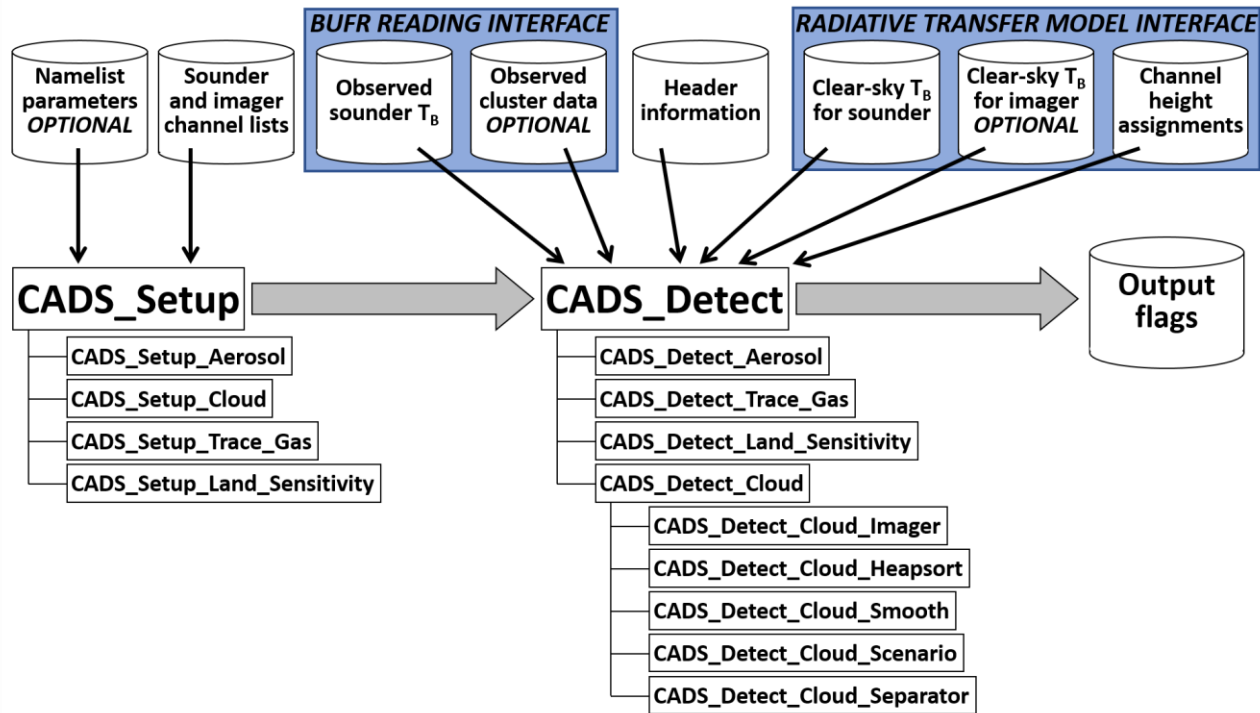


Figure 1: Functional elements and the data flow.

3.2 Functional elements

The overall code design is shown in Fig. 1. The output cloud, aerosol, trace-gas, and land-sensitivity flags are determined in a process that contains the following steps:

1. **CADS_Setup**: Read in tuning parameters values from the namelists and input channel lists. Allocate and initialize computing arrays accordingly. The main setup module is divided in four tasks corresponding to setting up computing structures for cloud, aerosol, trace gas, and land sensitivity detection:
 - **CADS_Setup_Aerosol**: Specify aerosol detection channels and tuning parameter values. Set up associated computing arrays.
 - **CADS_Setup_Cloud**: Specify cloud detection channels and tuning parameter values. Set up associated computing arrays.
 - **CADS_Setup_Trace_Gas**: Specify trace-gas detection channels and tuning parameter values. Set up associated computing arrays.
 - **CADS_Setup_Land_Sensitivity**: Specify tuning parameter values for the land sensitivity detection. Set up associated computing arrays.
2. **CADS_Detect**: Read in land fraction, channel height assignments, observed and calculated T_B data, and auxiliary information. Determine the cloud, aerosol, trace gas, and land-sensitivity flags by sequentially calling the four respective detection tasks:

The EUMETSAT Network of Satellite Application Facilities	 NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
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- **CADS_Detect_Aerosol:** Calculate inter-channel T_B differences across window channels and compare these against thresholds as defined in CADS_Setup_Aerosol. In the case of positive detection outcome, determine the aerosol type and AOD. Identify and flag affected channels.
- **CADS_Detect_Trace_Gas:** Calculate mean observed T_B and mean observed minus calculated T_B departure in a group of tracer channels and in another group of control channels. Comparing the tracer minus control group differences in observation and departure spaces, diagnose the presence / absence of excessive trace gas and raise the trace gas flags as necessary.
- **CADS_Detect_Land_Sensitivity:** Compare the input land fraction against its threshold as set in CADS_Setup_Land_Sensitivity. If the threshold is exceeded, normalize the input channel height assignments. Identify and flag those channels that have their normalized height assignment in excess of the threshold.
- **CADS_Detect_Cloud:** Identify cloud-affected channels. This task divides in five sub-tasks.
 - **CADS_Detect_Cloud_Imager:** Produce a preliminary FOV-specific cloud flag using the auxiliary information from a collocated imager (if this is available).
 - **CADS_Detect_Cloud_Heapsort:** Including only cloud detection channels as specified in CADS_Setup_Cloud, create an array of observed minus calculated T_B departure, and sort it according to the input channel height assignments.
 - **CADS_Detect_Cloud_Smooth:** Take the rank-sorted T_B departure array and smooth it by applying a moving average filter.
 - **CADS_Detect_Cloud_Scenario:** Depending on properties of the rank-sorted and smoothed T_B departure array, and the preliminary imager-based cloud flag when available, choose one of three possible cloud detection scenarios (Quick Exit, Warm Start, Cold Start).
 - **CADS_Detect_Cloud_Separator:** Taking the cloud detection scenario into account, study the rank-sorted and smoothed T_B departure array to find the index that separates cloud-affected and unaffected channels from each other. Using the height assignment of the lowest unaffected channel as the separator, flag all input channels either clear or cloudy.

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-DS-015 Version : v1.2 Date : 06/07/2020
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4 FUNCTIONAL REQUIREMENTS

Functional requirements for running the software include a FORTRAN 90 compiler and necessary external tools for providing the various inputs. The software itself handles ASCII files only. To obtain the software input data, user will need to implement interfaces to facilitate (a) reading observed T_B data and (b) running a radiative transfer software that produces both first-guess T_B and channel-specific height assignments. Most typically, the observed T_B data is read in from BUFR or HDF5 data, although it is usually necessary for the user to convert observation information from radiances to T_B . For radiative transfer calculations, the use of RTTOV is recommended.

Additionally, land fraction information needs to be provided by the user. This information can come from input BUFR data, or it can be interpolated to observation locations from a gridded land-sea mask product.